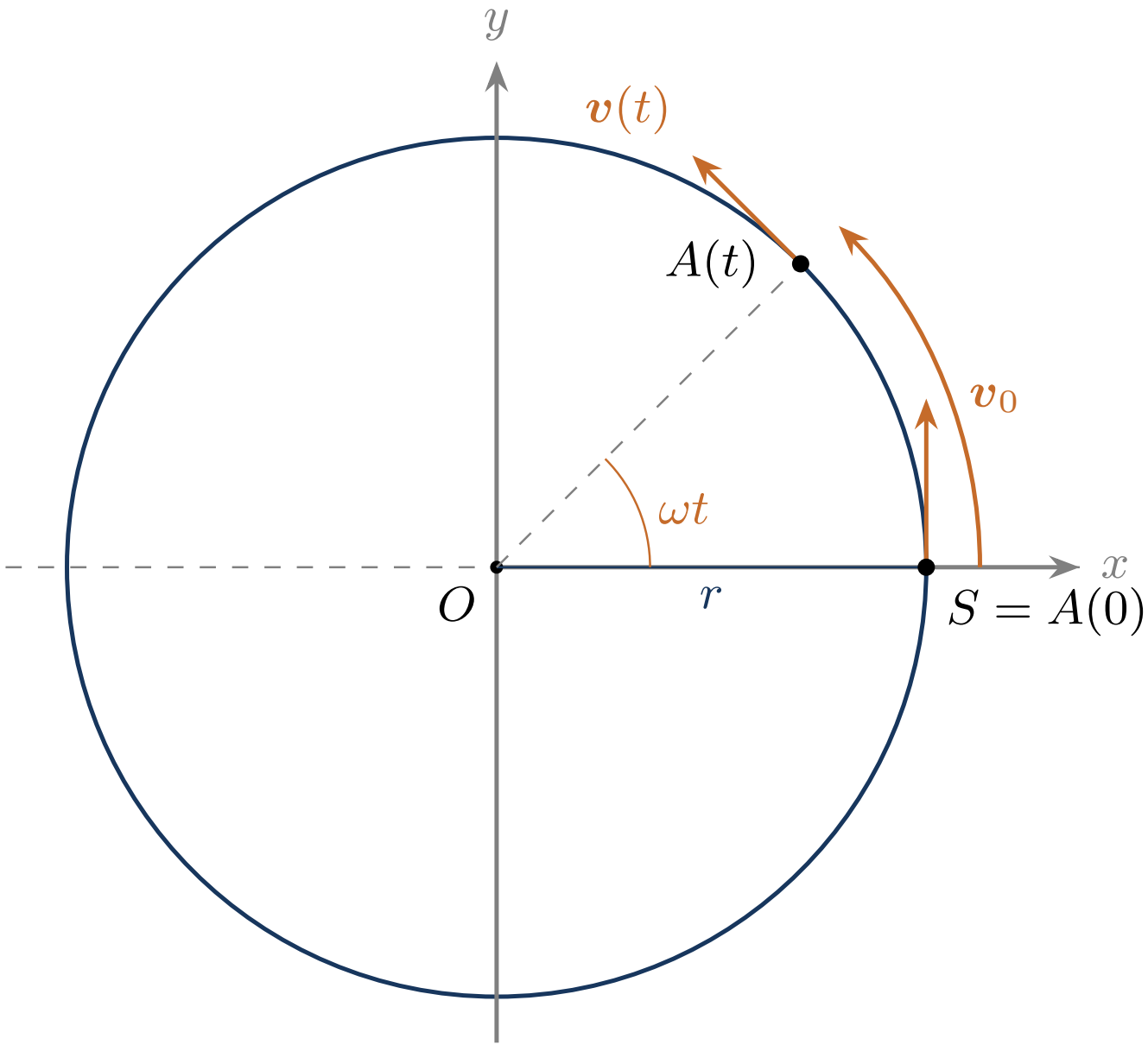
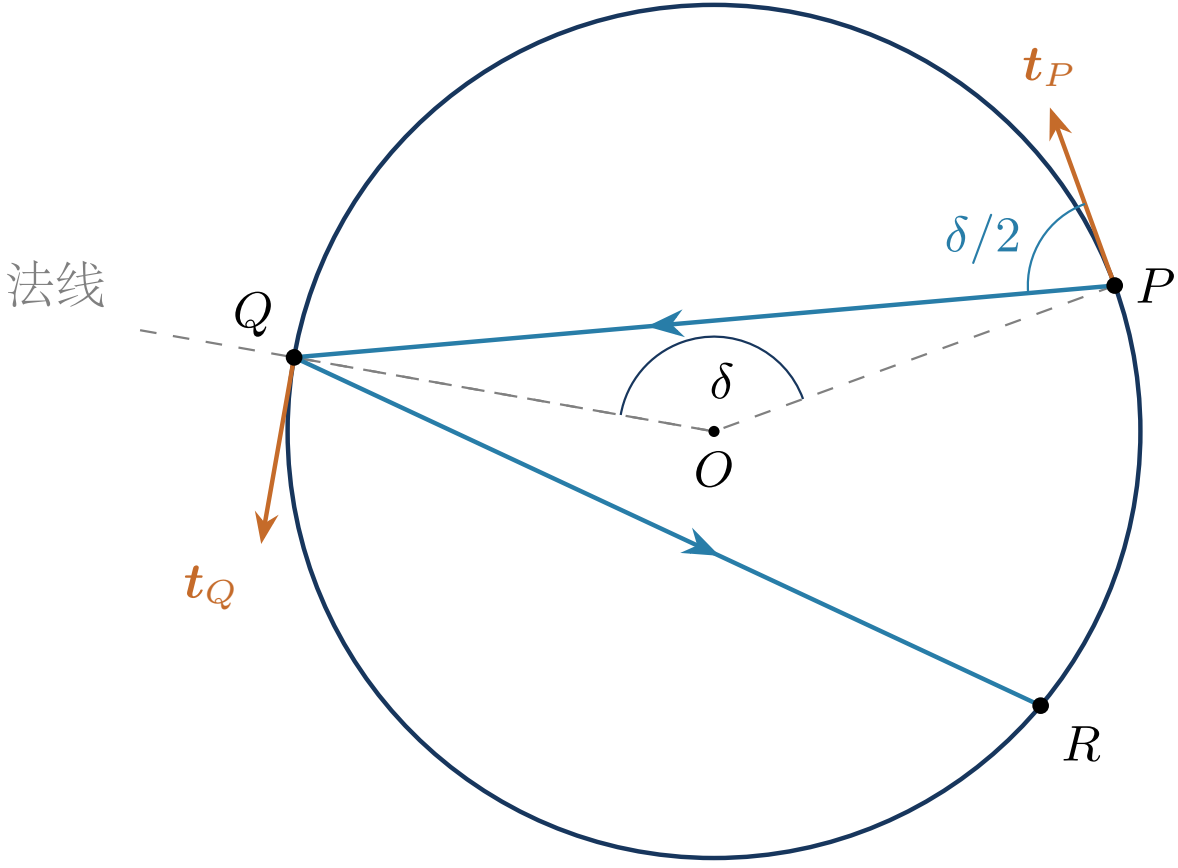


水平截面

空气静止，声速 c

跑动速度 $v = \omega r$

壁面固定，镜面反射



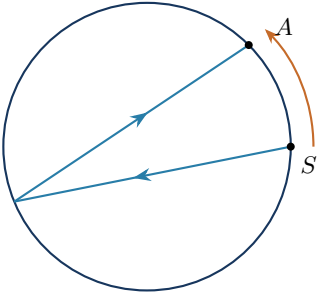
相邻反射点间的圆心角相同

$\angle POQ = \angle QOR = \delta$

图示为逆时针推进的声线

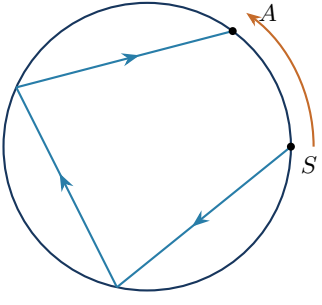
顺时针声线取相反切向投影

$d = 2r\sin(\delta/2)$



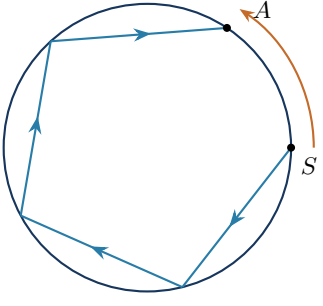
$k=2$，反射 1 次

$ct/r=3.923286$



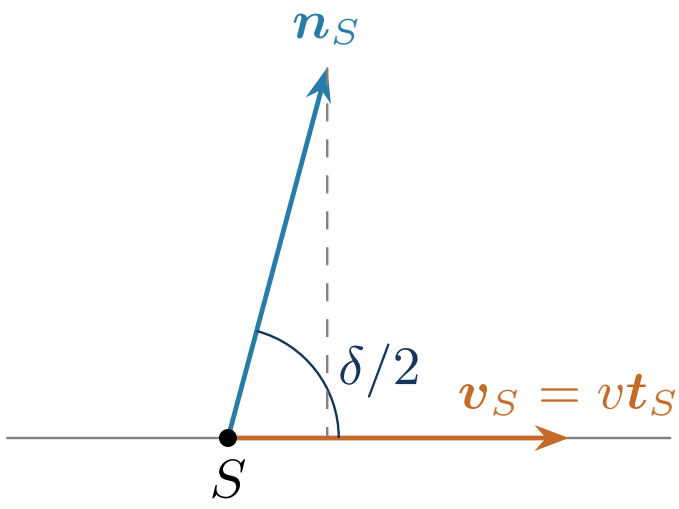
$k=3$，反射 2 次

$ct/r=4.668407$

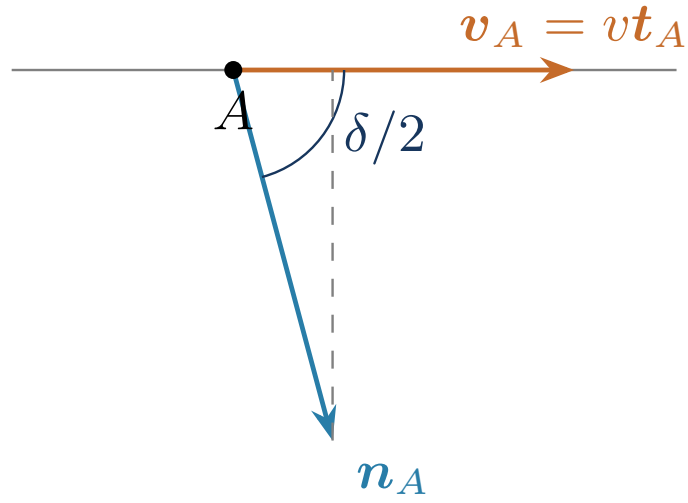


$k=4$，反射 3 次

$ct/r=4.920066$



发射端（沿声线方向投影）



接收端（已转至同向局部坐标）

$$\boldsymbol{v}_S \cdot \boldsymbol{n}_S = \boldsymbol{v}_A \cdot \boldsymbol{n}_A = +v\cos(\delta/2)$$

图示逆时针声线；顺时针声线两端均取负号